DATA SCIENCE LABORATORY MANUAL

Module I ? Data Science Fundamentals

Topic: Data Preprocessing

OVERVIEW & LEARNING OBJECTIVES

Data cleaning, imputation of missing entries, outlier mitigation, scaling, normalization, and one-hot encoding.

KEY TOPIC CONCEPTS & METHODOLOGY

- Understanding Data Preprocessing in modern Data Science workflows.
- Detailed breakdown: Data cleaning, imputation of missing entries, outlier mitigation, scaling, normalization, and one-hot encod
- Mathematical formulas, code examples, and laboratory execution guidelines.
- Common pitfalls, edge cases, and best practices in practical applications.

LABORATORY PRACTICE / IMPLEMENTATION NOTES

1. Follow standard Python and Data Science coding practices.
2. Document all exploratory findings and statistical hypotheses.
3. Keep datasets and outputs organized in your repository workspace.